

1.

$$1) A = \begin{pmatrix} 3 & 1 & 0 \\ 2 & 7 & 4 \\ 6 & 5 & 2 \end{pmatrix} \quad B = \begin{pmatrix} 4 & 2 & -3 \\ 5 & 7 & 0 \\ 0 & 0 & 1 \end{pmatrix} \quad A+B = \begin{pmatrix} 7 & 3 & -3 \\ 7 & 0 & 4 \\ 6 & 5 & 3 \end{pmatrix}$$

$$2) A = \begin{pmatrix} 2 & -1 \\ 3 & 0 \\ 5 & 8 \end{pmatrix} \quad B = \begin{pmatrix} 4 & 1 \\ -3 & -1 \\ 2 & 3 \end{pmatrix} \quad A+B = \begin{pmatrix} 6 & 0 \\ 0 & -1 \\ 7 & 11 \end{pmatrix}$$

$$3) A = \begin{pmatrix} 7 & 4 & 0 \\ -2 & 5 & 1 \end{pmatrix} \quad B = \begin{pmatrix} -7 & -4 & -1 \\ 2 & -5 & 0 \end{pmatrix} \quad A+B = \begin{pmatrix} 0 & 0 & -1 \\ 0 & 0 & 1 \end{pmatrix}$$

$$2. \quad 1) A = \begin{pmatrix} 2 & -1 & 4 \\ 0 & 5 & -3 \\ -2 & 1 & 0 \end{pmatrix}, k=3. \quad A \cdot k = \begin{pmatrix} 6 & -3 & 12 \\ 0 & 15 & -9 \\ -6 & 3 & 0 \end{pmatrix}$$

$$2) B = \begin{pmatrix} -2 & 0 & 4 & 5 \\ -7 & -3 & 10 & 0 \\ 1 & 9 & -1 & 0 \end{pmatrix}, k=-0,5 \quad B \cdot k = \begin{pmatrix} 1 & 0 & -2 & -2,5 \\ 3,5 & 1,5 & -5 & 0 \\ -0,5 & -4,5 & 0,5 & 0 \end{pmatrix}$$

$$3. \quad 1) 3A - 2B, \quad A = \begin{pmatrix} 2 & -4 & 0 \\ -1 & 5 & 1 \\ 0 & 3 & -7 \end{pmatrix} \quad B = \begin{pmatrix} 4 & -1 & -2 \\ 0 & -3 & 5 \\ 2 & 0 & -4 \end{pmatrix}$$

$$\begin{pmatrix} 3 \cdot 2 & 3 \cdot (-4) & 3 \cdot 0 \\ 3 \cdot (-1) & 3 \cdot 5 & 3 \cdot 1 \\ 3 \cdot 0 & 3 \cdot 3 & 3 \cdot (-7) \end{pmatrix} - \begin{pmatrix} 2 \cdot 4 & 2 \cdot (-1) & 2 \cdot (-2) \\ 2 \cdot 0 & 2 \cdot (-3) & 2 \cdot 5 \\ 2 \cdot 2 & 2 \cdot 0 & 2 \cdot (-4) \end{pmatrix} = \begin{pmatrix} 6 & -12 & 0 \\ -3 & 15 & 3 \\ 0 & 9 & -21 \end{pmatrix} - \begin{pmatrix} 8 & -2 & -4 \\ 0 & -6 & 10 \\ 4 & 0 & -8 \end{pmatrix} =$$

$$\begin{pmatrix} 6-8 & -12+2 & 0+4 \\ -3-0 & 15+6 & 3-10 \\ 0-4 & 9-0 & -21+8 \end{pmatrix} = \begin{pmatrix} -2 & -10 & 4 \\ -3 & 21 & -7 \\ -4 & 9 & -13 \end{pmatrix}$$

$$2) 2A + 3B - C = \begin{pmatrix} 2 \cdot 2 & 2 \cdot (-6) & 2 \cdot 1 \\ 2 \cdot 3 & 2 \cdot 0 & 2 \cdot 4 \end{pmatrix} + \begin{pmatrix} 3 \cdot (-5) & 3 \cdot 2 & 3 \cdot 3 \\ 3 \cdot 0 & 3 \cdot (-1) & 3 \cdot (-2) \end{pmatrix} - \begin{pmatrix} -1 & 5 & -3 \\ 0 & -7 & 9 \end{pmatrix} =$$

$$= \begin{pmatrix} 4+(-15) & -12+6 & 2+8 \\ 6+0 & 0+(-3) & 8+(-6) \end{pmatrix} - \begin{pmatrix} -1 & 5 & -3 \\ 0 & -7 & 9 \end{pmatrix} =$$

$$= \begin{pmatrix} -11+(-1) & -6-5 & 11-(-3) \\ 6-0 & -3-(-7) & 2-9 \end{pmatrix} = \begin{pmatrix} -10 & -11 & 14 \\ 6 & 4 & -7 \end{pmatrix}$$

4.

$$1) A = \begin{pmatrix} 3 & 2 & 1 \\ 0 & 1 & 2 \\ -3 & 1 & 1 \end{pmatrix}, B = \begin{pmatrix} -1 & 2 \\ 2 & 0 \\ -3 & 1 \end{pmatrix}, C = A \cdot B = \begin{pmatrix} -2 & 7 \\ -4 & 2 \end{pmatrix}$$

$$C_{11} = 3 \cdot (-1) + 2 \cdot 2 + 1 \cdot (-3) = -3 + 4 - 3 = -2$$

$$C_{12} = 3 \cdot 2 + 2 \cdot 0 + 1 \cdot 1 = 6 + 0 + 1 = 7$$

$$C_{21} = 0 \cdot (-1) + 1 \cdot 2 + 2 \cdot (-3) = 0 + 2 - 6 = -4$$

$$C_{22} = 0 \cdot 2 + 1 \cdot 0 + 2 \cdot 1 = 0 + 0 + 2 = 2$$

$$2) A = \begin{pmatrix} 3 & 0 \\ 0 & -1 \\ -2 & 5 \end{pmatrix}, B = \begin{pmatrix} -1 & 0 & 2 \\ 2 & 1 & 0 \end{pmatrix}, C = A \cdot B = \begin{pmatrix} -3 & 0 & 6 \\ -2 & -1 & 0 \\ 12 & 5 & -4 \end{pmatrix}$$

$$C_{11} = 3 \cdot (-1) + 0 \cdot 2 = -3$$

$$C_{12} = 3 \cdot 0 + 0 \cdot 1 = 0$$

$$C_{13} = 3 \cdot 2 + 0 \cdot 0 = 6$$

$$C_{21} = 0 \cdot (-1) + (-1) \cdot 2 = -2$$

$$C_{22} = 0 \cdot 0 + (-1) \cdot 1 = -1$$

$$C_{23} = 0 \cdot 2 + (-1) \cdot 0 = 0$$

$$C_{31} = (-2) \cdot (-1) + 5 \cdot 2 = 12$$

$$C_{32} = (-2) \cdot 0 + 5 \cdot 1 = 5$$

$$C_{33} = (-2) \cdot 2 + 5 \cdot 0 = -4$$

$$3) A = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}, B = \begin{pmatrix} 1 & 1 \\ -1 & -1 \end{pmatrix}, C = A \cdot B = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$$

$$C_{11} = 1 \cdot 1 + 1 \cdot (-1) = 0$$

$$C_{12} = 1 \cdot 1 + 1 \cdot (-1) = 0$$

$$C_{21} = 1 \cdot 1 + 1 \cdot (-1) = 0$$

$$C_{22} = 1 \cdot 1 + 1 \cdot (-1) = 0$$

$$4) A = \begin{pmatrix} 3 & 1 & 0 \\ 2 & 7 & 4 \\ 6 & 5 & 2 \end{pmatrix}, B = \begin{pmatrix} 4 & 2 & -3 \\ 5 & 7 & 0 \\ 0 & 0 & 1 \end{pmatrix}, C = A \cdot B = \begin{pmatrix} 17 & 13 & -9 \\ -27 & -45 & -2 \\ 49 & 47 & -16 \end{pmatrix}$$

$$C_{11} = 3 \cdot 4 + 1 \cdot 5 + 0 \cdot 0 = 12 + 5 = 17$$

$$C_{12} = 3 \cdot 2 + 1 \cdot 7 + 0 \cdot 0 = 6 + 7 = 13$$

$$C_{13} = 3 \cdot (-3) + 1 \cdot 0 + 0 \cdot 1 = -9$$

$$C_{21} = 2 \cdot 4 + 7 \cdot 5 + 4 \cdot 0 = 8 + 35 = 43$$

$$C_{22} = 2 \cdot 2 + 7 \cdot 7 + 4 \cdot 0 = 4 + 49 = 53$$

$$C_{23} = 2 \cdot (-3) + 7 \cdot 0 + 4 \cdot 1 = -6 + 0 + 4 = -2$$

$$C_{31} = 6 \cdot 4 + 5 \cdot 5 + 2 \cdot 0 = 24 + 25 = 49$$

$$C_{32} = 6 \cdot 2 + 5 \cdot 7 + 2 \cdot 0 = 12 + 35 = 47$$

$$C_{33} = 6 \cdot (-3) + 5 \cdot 0 + 2 \cdot 1 = -18 + 2 = -16$$